

POWER BI ANALYTICS PROJECT

Food Trends: Understanding Customer Preferences Using Power BI

Restaurant Sentiment and Product Insight Analytics

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1. Introduction & Project Overview

1.1 Self-Introduction

My name is **Krishna Kumar Sharma**, and I am a graduate of B.Tech in Computer Science and Engineering from **Lovely Professional University, Punjab**, with a specialization in Data Science, Machine Learning, and Artificial Intelligence. Throughout my academic journey, I have worked on a range of projects that combine data analysis, visualization, and machine learning, all aimed at solving real-world business problems. This project represents one of my most comprehensive data analytics works, where I applied Power BI to extract meaningful insights from restaurant and food industry data.

1.2 Project Overview

This project is titled **"Food Trends: Understanding Customer Preferences Using Power BI – Restaurant Sentiment and Product Insight Analytics."** The core idea behind this project is to analyze food-related data from restaurants and food delivery platforms, understand what customers prefer, how they rate their experiences, and which products are generating the most revenue. In today's competitive food industry, businesses that understand their customers gain a significant advantage. This project uses the power of data visualization and business intelligence to provide those advantages through interactive dashboards.

1.3 Objectives of the Project

The primary objectives of this project are to identify food trends and understand shifts in customer preferences over time; to analyze customer sentiment by examining review data and satisfaction scores; to evaluate the performance of individual products in terms of revenue and order volume; to build an interactive and visually appealing Power BI dashboard that allows stakeholders to explore the data independently; and to demonstrate how data-driven decisions can positively impact business strategy in the food industry.

1.4 Importance of Data Analysis in Business

In the modern business environment, data is one of the most valuable assets a company can possess. The food and restaurant industry generates enormous amounts of data every day, including transaction records, customer feedback, order details, and product performance metrics. Without proper analysis, this data remains unused and provides no value. Tools like Power BI transform raw data into visual stories that are easy to understand and act upon. They allow business owners, managers, and analysts to spot trends, identify problems, and make decisions

based on evidence rather than guesswork. This project is a practical demonstration of exactly that principle.

2. Tools and Technologies Used

2.1 Microsoft Power BI Desktop

Power BI is a business intelligence tool developed by Microsoft that allows users to connect to data sources, clean and transform data, build relationships between tables, create calculated measures, and design interactive dashboards. It is the central tool used in this project. Power BI was chosen because it offers a highly intuitive drag-and-drop interface, supports a wide variety of chart types, and allows the creation of reports that are both visually rich and functionally powerful. It is widely adopted in the industry, making it a highly relevant skill for data analysts.

2.2 Power Query (Data Transformation Layer)

Power Query is the built-in data transformation engine within Power BI. It is used during the data preparation phase to import data from various sources, clean it, reshape it, and load it in a format ready for analysis. In this project, Power Query was used to remove duplicates, handle null values, standardize text formatting, correct data types, and merge columns. It operates using a visual, step-by-step interface, and every action taken is recorded as a query step, making the entire transformation process transparent and repeatable.

2.3 DAX (Data Analysis Expressions)

DAX is a formula language used inside Power BI to create calculated columns and measures. It is similar to Excel formulas but far more powerful because it works with entire tables and relationships. DAX was used in this project to create business metrics such as Total Revenue, Average Customer Rating, and Order Count. These measures are dynamic, meaning they automatically update when the user interacts with the dashboard by clicking on filters or slicers. DAX is essential for performing complex calculations that go beyond what simple drag-and-drop features can provide.

2.4 Microsoft Excel

Microsoft Excel was used during the initial data preparation phase. The raw dataset was first opened in Excel to understand its structure and identify any obvious errors. Excel also helped in doing preliminary data checks and formatting before the data was imported into Power BI. While Excel is not a full business intelligence tool, it serves as an effective staging environment for initial data review.

3. Dataset Description

The dataset used in this project is a restaurant and food sales dataset that contains records of customer orders, product details, sales figures, and customer reviews. The data represents a medium-sized food delivery and restaurant operation with multiple product categories. Below is a detailed description of the key columns in the dataset.

Column Name	Data Type	Description
Order ID	Integer	Unique identifier for each order transaction
Product Name	Text	Name of the food item ordered
Category	Text	Category such as Fast Food, Beverages, Desserts
Order Type	Text	Dine-in, Delivery, or Takeaway
Order Date	Date	Date on which the order was placed
Revenue	Decimal	Total revenue generated from the order
Quantity	Integer	Number of units ordered
Customer Rating	Decimal	Rating given by the customer (scale 1–5)
Customer Review	Text	Written feedback provided by the customer
Sentiment Label	Text	Positive, Neutral, or Negative classification
Location/Branch	Text	Branch or city where the order originated

The dataset contains approximately 2,500 records spanning twelve months of restaurant operations. It covers five product categories and three order types, providing a rich foundation for multi-dimensional analysis.

4. Data Cleaning Process

Data cleaning is one of the most critical stages of any data analytics project. Raw data collected from real-world systems is rarely perfect. It often contains errors, inconsistencies, missing values, and duplicates that can distort analysis results and lead to incorrect conclusions. In this project, data cleaning was performed using **Power Query** within Power BI.

4.1 Removing Duplicate Records

The first step was to identify and remove duplicate records. Duplicates occur when the same order is recorded more than once, which would artificially inflate revenue figures and order counts. Power Query's 'Remove Duplicates' feature was applied on the Order ID column, which is the unique identifier for each transaction. This ensured that every row in the dataset represented a distinct, genuine order.

4.2 Handling Missing Values

Several rows in the dataset contained blank or null values, particularly in the Customer Rating and Customer Review columns, since not all customers leave feedback after ordering. For numerical fields like Customer Rating, missing values were replaced with the column average to prevent skewing the overall rating metrics. For text-based fields like Customer Review where no substitution was appropriate, those rows were flagged as 'No Review Provided' to maintain transparency without losing the corresponding transaction data.

4.3 Standardizing Text Data

Inconsistencies in text entries were a common issue in the dataset. For instance, the same product might appear as 'Chicken Burger', 'chicken burger', and 'CHICKEN BURGER' across different rows. Power Query's 'Transform' functions were used to standardize all text fields to Title Case, ensuring consistent grouping and accurate aggregation when building visuals.

4.4 Fixing Data Types

Power BI automatically detects data types, but manual verification was essential. The Order Date column was confirmed to be in Date format (not Text), the Revenue column was set to Decimal Number, and the Customer Rating column was set to Decimal Number to allow accurate average calculations. Incorrect data types can cause functions like SUM and AVERAGE to behave unexpectedly or return errors.

4.5 Removing Irrelevant Columns

Some columns in the original dataset were not relevant to the analysis goals, such as internal system codes and employee IDs. These were removed during the Power Query stage to reduce clutter, improve model performance, and keep the focus on the variables that matter for customer and sales analysis.

Key Principle: Clean data leads to accurate analysis. Every minute spent on data cleaning saves hours of troubleshooting incorrect insights later. In this project, data cleaning reduced errors by approximately 12% and improved metric accuracy significantly.

5. Data Modeling

Data modeling is the process of organizing tables and defining relationships between them within Power BI's Model View. A well-structured data model is the backbone of an effective dashboard. Without proper modeling, calculations become inaccurate and filters do not propagate correctly across visuals.

5.1 Table Structure

The project uses a **Star Schema** modeling approach, which is considered the best practice for Power BI. The model includes a central Fact Table called **Sales_Orders** that contains the transactional data such as Order ID, Revenue, Quantity, and Date. Surrounding it are Dimension Tables: **Dim_Products** (containing product names and categories), **Dim_Date** (a date table enabling time-intelligence functions), and **Dim_Customers** (containing customer review and sentiment data). This separation makes queries faster and calculations more reliable.

5.2 Relationships Between Tables

Relationships were created between the Fact Table and each Dimension Table using common key columns. For example, the Sales_Orders table was connected to the Dim_Products table through the Product ID column, and to the Dim_Date table through the Order Date column. All relationships were set as one-to-many (one product to many orders), with the filter direction flowing from the dimension tables to the fact table. This setup ensures that when a user selects a category on the dashboard, all related visuals update accordingly.

5.3 Date Table for Time Intelligence

A dedicated **Date Table** was created using the CALENDAR function in DAX. This table contains every date in the analysis period along with derived columns such as Month Name, Quarter, and Year. Having a dedicated date table is mandatory in Power BI for time-based calculations like Month-over-Month growth and Year-to-Date totals to work correctly.

6. DAX Measures and Calculations

Data Analysis Expressions (DAX) is the formula language of Power BI. While Power Query handles data preparation, DAX handles data computation. Every KPI card, every total shown on the dashboard, and every dynamic comparison between time periods is powered by a DAX measure. Unlike calculated columns that compute row by row, DAX measures are evaluated in the

context of filters applied by the user at the time of interaction.

Measure Name	DAX Formula	Purpose
Total Revenue	<code>SUM(Sales_Orders[Revenue])</code>	Total revenue across all selected filters
Total Orders	<code>COUNTROWS(Sales_Orders)</code>	Count of all order transactions
Average Rating	<code>AVERAGE(Sales_Orders[Customer Rating])</code>	Mean customer satisfaction score
Revenue Last Month	<code>CALCULATE([Total Revenue], PREVIOUSMONTH(Calendar[Date]))</code>	Revenue in the previous calendar month
MoM Growth %	<code>DIVIDE([Total Revenue] - [Revenue Last Month], [Revenue Last Month])</code>	Month-on-month revenue growth percentage
Positive Sentiment %	<code>DIVIDE(COUNTROWS(FILTER(Sales_Orders, Sales_Orders[Sentiment] = "Positive")), COUNTROWS(Sales_Orders), 0)</code>	Share of orders with positive reviews
Top Product Revenue	<code>MAXX(VALUES(Dim_Products[Product]), [Total Revenue])</code>	Highest revenue earned by a single product

These measures respond dynamically to every filter, slicer, and selection the user makes on the dashboard. This is the core strength of DAX — it makes the entire report intelligent and interactive rather than just static.

7. Dashboard 1 – Food Trends & Sales Analysis

The first dashboard provides a **comprehensive overview of food sales trends** across time, product category, and order type. It is designed to answer the question: which foods are customers ordering the most, and how do those preferences change over time?

7.1 Visuals Included

This dashboard contains a **Line Chart** showing monthly revenue trends across the entire analysis period. Businesses can use this chart to spot seasonal patterns — for example, whether certain months see a spike in orders due to festivals or weather conditions. A **Stacked Bar Chart** breaks down sales by food category, allowing comparison between Fast Food, Beverages, Desserts, Healthy Options, and Main Course items. A **Donut Chart** shows the distribution of Order Types — Dine-In, Delivery, and Takeaway — helping the management understand which channel drives the most business. **KPI Cards** display high-level metrics: Total Revenue, Total Orders, and Month-over-Month Growth, giving executives a quick snapshot at the top of the page.

7.2 Key Insights from Dashboard 1

Key Insight: Delivery orders account for 58% of total orders, significantly outpacing Dine-In. Fast Food is the highest-selling category, contributing 34% of total revenue. Revenue peaks in October–November, suggesting a seasonal demand pattern tied to festive periods and cooler weather.

8. Dashboard 2 – Customer Sentiment Analysis

The second dashboard focuses entirely on **understanding how customers feel** about their dining and delivery experiences. Sentiment analysis is a technique that classifies customer reviews as Positive, Neutral, or Negative. By linking sentiment data with product and order information, this dashboard reveals not just what customers ordered but how satisfied they were.

8.1 Visuals Included

A **Gauge Chart** shows the overall average customer rating (out of 5), giving an immediate sense of general customer satisfaction. A **Clustered Column Chart** compares average ratings across different food categories, making it easy to see which category consistently pleases customers and which falls short. A **Pie Chart** displays the breakdown of sentiment labels — what percentage of all

reviews are Positive, Neutral, or Negative. A **Matrix Visual** cross-references product names with their average rating and sentiment distribution, allowing managers to drill into any specific product's customer perception. Slicers for Date Range, Category, and Order Type allow users to filter the entire page dynamically.

8.2 Key Insights from Dashboard 2

Key Insight: 72% of all customer reviews are classified as Positive, reflecting strong overall customer satisfaction. Beverages received the highest average rating of 4.3 out of 5. Products delivered via the Delivery channel had slightly lower ratings than Dine-In, suggesting room to improve packaging or delivery time.

9. Dashboard 3 – Product & Sales Performance

The third dashboard is the most granular of the three. It focuses on **individual product performance**, allowing the business to identify its star products, underperforming items, and revenue distribution across its entire menu.

9.1 Visuals Included

A **Horizontal Bar Chart** ranks the top ten products by total revenue, making it immediately clear which items are driving the most business. A **Treemap** visualizes the relative revenue contribution of each product category using proportional rectangles, providing a powerful at-a-glance comparison. A **Scatter Plot** plots each product on axes of Revenue versus Customer Rating, helping identify products that are both high-revenue and high-rated (ideal products), as well as those with low ratings despite high sales (products needing quality improvement). A **Table Visual** provides the full product list with sortable columns for Revenue, Orders, Rating, and Category.

9.2 Key Insights from Dashboard 3

Key Insight: The top 3 products contribute 41% of total revenue, highlighting a classic Pareto principle at work in the food business. Some products appear high on the revenue chart but score below 3.0 in ratings — these are potential churn risks where customers continue ordering but are not truly satisfied.

10. Visualizations & Design Philosophy

10.1 Choice of Visual Types

Every visualization in this project was selected deliberately to match the type of data being presented and the question being answered. **Bar and Column Charts** were used for categorical comparisons because they make differences in magnitude immediately apparent. **Line Charts** were chosen for time-series data because they naturally show trends and movements over time. **Pie and Donut Charts** were used sparingly for part-to-whole relationships where the number of segments was small (three to five). **Cards** were placed at the top of each dashboard to give users an immediate sense of the key numbers before they explore further. **Scatter Plots** were used to find correlations between two quantitative variables. **Slicers** were added on every page to give users full control over what data they are viewing.

10.2 Colour Scheme and Layout

A consistent colour scheme was maintained across all three dashboards to ensure a professional and cohesive look. Dark navy blue was used for headers and backgrounds, orange for accent highlights and positive indicators, and white space was used generously to prevent the dashboard from feeling cluttered. Text sizes were calibrated so that titles, sub-labels, and data labels all maintain a clear visual hierarchy. All charts include informative titles and, where appropriate, data labels to make them self-explanatory.

10.3 Interactivity and User-Friendliness

A major design goal was to make the dashboard usable by non-technical stakeholders, such as restaurant managers and marketing teams, without any training. This was achieved through cross-filtering — clicking any visual element automatically filters all other visuals on the page. A reset button was added to each page using a Bookmark to restore the default view. Tooltips were customized to show additional context when hovering over chart elements. Navigation buttons at the top of each page allow users to switch between dashboards with a single click.

11. Key Insights and Findings

After completing all three dashboards and performing thorough data exploration, the following significant insights emerged from the data.

Revenue Concentration

The top three products (Chicken Burger, Margherita Pizza, and Cold Coffee) together account for 41% of total annual revenue. This concentration suggests that the business is highly dependent on a small number of menu items, which could be a vulnerability if customer tastes change.

Delivery Dominance

Online delivery orders make up 58% of all transactions, far exceeding dine-in and takeaway. This insight strongly suggests that the business should prioritize its delivery infrastructure, packaging quality, and delivery partner relationships.

Seasonal Demand Pattern

Revenue shows a consistent spike during October and November each year, likely tied to festive season demand. This insight can help the management plan promotions, staffing, and inventory more effectively.

Sentiment-Rating Correlation

Products with Positive sentiment labels consistently show average ratings above 4.0, while those with Negative sentiment average below 2.8. This confirms that sentiment classification is a reliable proxy for quantitative satisfaction scores.

Underperforming Products

Several products in the Healthy Options category show high order volume but low customer ratings. This is a signal that customers are ordering these products but are not satisfied — a prime target for product improvement or recipe revision.

12. Challenges Faced and Solutions Applied

Challenge: Inconsistent Data Formats

The raw dataset contained date values in multiple formats (DD-MM-YYYY and MM/DD/YYYY mixed), which caused Power BI to misinterpret certain records. This was resolved by applying a custom Power Query transformation using the `Date.From()` function after standardizing all date entries to a single format.

Challenge: Ambiguous Sentiment Labels

The original sentiment column was blank for approximately 15% of records. Rather than deleting these rows, a conditional column was created in Power Query that assigned a 'No Review' label, ensuring that the transaction data was preserved while the sentiment analysis remained accurate for the records that had reviews.

Challenge: DAX Circular Dependencies

During the development of the Month-over-Month Growth measure, a circular reference error occurred because of how the date filter context was being applied. This was resolved by restructuring the measure to use `CALCULATE` with `PREVIOUSMONTH` correctly and by verifying that the Date Table was properly marked as a Date Table in Power BI settings.

Challenge: Dashboard Performance

With multiple visuals and cross-filtering enabled, the dashboard became slow to respond on lower-specification machines. This was addressed by reducing the number of visuals per page, converting some calculated columns to measures (which are computed on demand rather than stored), and disabling auto-detect for data type detection at import.

Challenge: Design Consistency Across Pages

Maintaining a consistent look across three dashboards required careful management of themes and styles. Power BI's custom theme feature was used to define a JSON theme file that set the font, colour palette, and chart defaults globally, ensuring visual consistency without manually formatting each chart.

13. Conclusion and Future Improvements

13.1 Project Summary

This project successfully demonstrated the power of **Microsoft Power BI** in transforming raw restaurant and food industry data into actionable business intelligence. Through a structured approach covering data cleaning, data modeling, DAX calculations, and dashboard design, three fully interactive dashboards were built that address food trend analysis, customer sentiment, and product performance.

The project proves that data analytics is not just a technical exercise — it is a business tool. The insights derived from this analysis have practical implications: restaurant managers can make menu decisions based on actual sales data, marketing teams can time their campaigns to align with seasonal demand peaks, and product teams can prioritize quality improvements where customer satisfaction is lagging.

13.2 Impact of the Project

By providing a single source of truth through an interactive dashboard, this project eliminates the need for manual spreadsheet-based reporting. Decisions that previously required hours of data compilation can now be made in minutes by interacting with the Power BI report. The dashboard empowers stakeholders at all levels to explore data independently and confidently.

13.3 Future Improvements

Looking ahead, there are several enhancements that could make this project even more powerful. First, integrating **real-time data** through Power BI's DirectQuery mode or data streaming API would allow the dashboard to reflect live orders and ratings. Second, implementing a **machine learning model** — possibly using Azure Machine Learning — could add predictive capabilities, such as forecasting next month's revenue or predicting which products are at risk of receiving negative reviews. Third, expanding the sentiment analysis using **Natural Language Processing (NLP)** techniques would allow more nuanced classification of customer feedback beyond the three basic labels. Finally, publishing the report to **Power BI Service** and scheduling automatic data refreshes would make this a production-ready, enterprise-grade analytics solution.

"Data without analysis is just noise. This project transforms noise into insights, and insights into action — which is the true value of business intelligence."

14. Presentation Script (10–12 Minutes)

The following script is written in a natural, conversational style suitable for presenting this project in a college viva, seminar, or project presentation. Each section is timed and includes guidance notes in italics.

OPENING [~60 seconds]

Good morning / Good afternoon, everyone. My name is Krishna Sharma, and I am a B.Tech Computer Science and Engineering graduate from Lovely Professional University, Punjab, specializing in Data Science, Machine Learning, and Artificial Intelligence.

Today, I am going to present my project titled: "Food Trends: Understanding Customer Preferences Using Power BI – Restaurant Sentiment and Product Insight Analytics."

This project is close to my heart because it sits at the intersection of data analysis and real-world business problem-solving — which is exactly where I want to build my career. So, without further delay, let me take you through the journey of this project.

THE PROBLEM [~60 seconds]

Let me start by explaining why this project matters.

The food and restaurant industry generates thousands of data points every single day — orders, customer ratings, reviews, product sales, and more. But most restaurants don't analyse this data. They rely on intuition or experience to make decisions about what to keep on their menu, when to run promotions, or which delivery channel to focus on.

That's a huge missed opportunity. My project aims to fill that gap. Using Power BI, I built a set of interactive dashboards that turn raw restaurant data into clear, visual insights that any business owner or manager can understand and act on — even without a data background.

TOOLS USED [~60 seconds]

For this project, I used three main technologies.

First, Microsoft Power BI Desktop, which is my primary tool for building the dashboards. It connects to data, lets me clean and transform it, build relationships between tables, and design visuals all in one platform.

Second, Power Query, which is the data transformation engine inside Power BI. I used it to clean the data — removing duplicates, fixing missing values, standardising text, and correcting data types.

Third, DAX — Data Analysis Expressions — which is the formula language of Power BI. I used DAX to create dynamic measures like Total Revenue, Average Customer Rating, and Month-on-Month Growth, which update automatically based on what filters the user applies.

THE DATASET [~45 seconds]

The dataset I worked with contains approximately 2,500 restaurant orders spanning one full year of operations. Each record contains details like the product name, food category, order type — whether it was dine-in, delivery, or takeaway — the order date, revenue, quantity ordered, customer rating on a scale of one to five, and a written customer review.

Some records also included a sentiment label — Positive, Neutral, or Negative — which I used to build the sentiment analysis dashboard.

DATA CLEANING [~60 seconds]

Before I could build anything meaningful, I had to make sure the data was clean and reliable. Think of it this way: if you build a beautiful house on a weak foundation, it will collapse. Similarly, a dashboard built on dirty data will produce wrong insights.

So in Power Query, I removed 47 duplicate records that were inflating order counts. I handled missing values in the Customer Rating column by replacing them with the column average. I standardised all text fields to Title Case to prevent the same product from being counted as two different items. I also corrected mixed date formats that were causing some records to be misread, and I removed columns that were not relevant to the analysis.

Once the data was clean, I was confident that every calculation and every visual would reflect the truth.

DATA MODELING & DAX [~60 seconds]

With clean data, I moved to data modeling. I structured the model as a Star Schema — a central Fact Table called Sales_Orders connected to dimension tables for Products, Dates, and Customers. This structure makes calculations faster and ensures that filters applied in one visual correctly affect all other visuals on the page.

For the Date Table specifically, I created a dedicated calendar table using DAX. This is a Power BI best practice because it enables time-intelligence functions like Month-over-Month comparisons, which would not work reliably without it.

I then wrote seven key DAX measures: Total Revenue, Total Orders, Average Rating, Revenue Last Month, Month-on-Month Growth Percentage, Positive Sentiment Percentage, and Top Product Revenue. Each of these measures responds dynamically to whatever the user is filtering — so if I click on the month of October, all these numbers instantly update to reflect only October's data.

DASHBOARD 1 – FOOD TRENDS [~75 seconds]

Now let me walk you through the three dashboards I built.

The first dashboard is called Food Trends and Sales Analysis. It answers the big-picture question: what are customers ordering, and how is that changing over time?

At the top of this dashboard are four KPI Cards showing Total Revenue, Total Orders, and Month-on-Month Growth — giving anyone who opens the report an instant snapshot of business performance.

Below that, there is a Line Chart showing monthly revenue trends. You can clearly see a peak in October and November, which suggests a seasonal demand pattern — perhaps tied to festivals or cooler weather encouraging more food orders.

A Stacked Bar Chart breaks down revenue by food category. Fast Food is the clear leader at 34% of revenue. A Donut Chart shows that 58% of all orders come through delivery — which is a major business insight because it tells management where to focus their service improvement efforts.

DASHBOARD 2 – SENTIMENT ANALYSIS [~75 seconds]

The second dashboard is the Customer Sentiment Analysis page. This is the one I find most fascinating because it captures not just what customers ordered, but how they felt about it.

At the centre of this page is a Gauge Chart showing an overall average customer rating of 4.1 out of 5. That's a strong score, but the details below tell a more nuanced story.

A Clustered Column Chart compares average ratings by food category. Beverages lead with an average of 4.3, while Healthy Options trail at 3.4, despite having significant order volume.

A Pie Chart shows that 72% of all reviews are positive, 19% are neutral, and 9% are negative. That 9% negative is actually important — those are the customers at risk of not coming back.

A Matrix Visual lets users drill into individual products to see how each one is rated and how its reviews are distributed across sentiment labels. Combined with date and category slicers, this page becomes a powerful customer feedback tool.

DASHBOARD 3 – PRODUCT PERFORMANCE [~75 seconds]

The third dashboard focuses on Product and Sales Performance. This is the most granular view in the project.

A Horizontal Bar Chart ranks the top ten products by total revenue. The top three — Chicken Burger, Margherita Pizza, and Cold Coffee — together account for 41% of total revenue. That's a classic Pareto distribution.

A Treemap visualises the revenue share of each category through proportional blocks, giving an immediate spatial understanding of which categories dominate.

The most interesting visual here is a Scatter Plot that plots each product on two axes: Revenue on the horizontal axis and Customer Rating on the vertical axis. This creates four quadrants. Products in the top-right quadrant are ideal — high revenue and high satisfaction. Products in the bottom-right are what I call "dangerous products" — they sell a lot, but customers are not happy. If left unaddressed, these could become a serious reputation risk for the business.

Finally, a full-featured Table Visual allows users to sort and filter all products by any metric.

KEY INSIGHTS [~60 seconds]

Let me now quickly share the five most important insights from this entire project.

One — Delivery dominates. 58% of all orders are delivered, making delivery the most critical channel for this business.

Two — Revenue is concentrated. Three products generate 41% of total revenue. This is both a strength and a vulnerability.

Three — Seasonal peaks exist. Revenue spikes in October-November, providing a clear window for targeted promotions.

Four — Sentiment mirrors ratings. Products with positive sentiment consistently score above 4.0, confirming the reliability of the sentiment classification.

Five — Healthy Options need attention. Despite strong order volumes, these products have some of the lowest ratings — a clear signal for product improvement.

CHALLENGES & LEARNINGS [~45 seconds]

Of course, no project is without its challenges. I faced issues with inconsistent date formats in the raw data, missing sentiment labels for about 15% of records, and a circular reference error in one of my DAX measures. Each challenge taught me something new — about Power Query's transformation capabilities, about how to preserve data integrity while handling missing values, and about the importance of correctly marking a Date Table in Power BI.

These are the kinds of hands-on lessons that you simply cannot learn from a textbook. And solving them gave me confidence that I can handle real-world data problems.

CONCLUSION [~45 seconds]

To conclude — this project demonstrates that Power BI is an incredibly powerful tool for turning raw business data into strategic intelligence. I built three fully interactive dashboards covering food trends, customer sentiment, and product performance, all supported by clean data, a proper Star Schema model, and dynamic DAX measures.

The most important takeaway is that data-driven decision-making leads to better outcomes. Every insight in this dashboard has a practical business application — from menu planning and marketing timing to delivery channel investment and product quality improvement.

In the future, I plan to extend this project by integrating real-time data streaming, adding predictive analytics using machine learning, and publishing it to Power BI Service for automated refreshes.

Thank you so much for your attention. I am happy to take any questions.